

PRETORIA EENDRACHT, South Africa

WMO: 682620

Lat: 25.7426S

Lon: 28.1857E

Elev: 4291

StdP: 12.56

Time Zone: 2.00 (E02)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.0	39.0	12.7	12.3	59.4	16.2	14.6	59.2	12.1	50.8	11.0	53.2	2.5	330	0.290

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	18.0	90.0	63.2	88.0	63.1	86.1	63.2	69.9	80.8	68.9	79.5	68.0	78.3	4.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	116.7	72.8	66.0	112.7	72.5	64.9	108.3	72.1	36.8	80.8	35.9	79.5	35.1	78.4	73.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
11.6	9.8	8.7	DB	32.7	95.3	2.3	2.6	31.0	97.1	29.7	98.6	28.4	100.1	26.7	101.9
			WB	28.1	71.0	2.2	1.5	26.5	72.1	25.3	73.0	24.0	73.9	22.4	75.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.3	73.4	73.7	71.2	66.0	60.0	55.1	54.1	60.3	67.4	70.3	71.6	72.5	(6)
	DBStd	8.24	4.18	3.81	4.21	4.30	4.20	3.66	3.91	5.26	5.66	5.55	5.37	4.06	(7)
	HDD50	19	0	0	0	0	1	6	10	2	0	0	0	0	(8)
	HDD65	1077	2	0	6	39	160	298	337	164	38	18	12	3	(9)
	CDD50	5953	725	662	657	480	312	158	138	321	523	629	649	699	(10)
	CDD65	1537	262	243	198	69	7	0	0	18	111	181	211	237	(11)
	CDH74	10762	1775	1573	1157	438	82	6	12	211	970	1453	1559	1526	(12)
	CDH80	3043	536	467	295	75	6	0	1	24	266	473	496	404	(13)
(14) Wind	WSAvg	3.3	3.6	3.2	3.0	2.7	2.4	2.7	2.9	3.5	3.9	4.1	4.2	3.8	(14)
(15) Precipitation	PrecAvg	26.8	5.2	3.3	3.2	2.1	0.5	0.1	0.1	0.2	0.7	3.2	4.1	4.4	(15)
	PrecMax	37.3	19.8	8.4	7.6	5.8	2.4	0.9	1.0	1.0	3.0	8.9	8.5	8.0	(16)
	PrecMin	16.0	1.7	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.6	0.8	1.3	(17)
	PrecStd	5.1	3.3	1.9	2.0	1.8	0.7	0.3	0.2	0.2	0.7	2.1	2.2	1.7	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	92.1	91.5	90.0	86.1	81.0	74.6	75.4	83.0	89.8	92.0	92.0	89.9	(19)
		MCWB	66.0	65.3	64.0	61.2	57.3	53.4	51.1	55.4	58.5	60.1	62.2	64.9	(20)
	2%	DB	88.2	88.0	86.2	82.1	76.6	71.4	71.9	79.1	86.3	88.3	88.4	87.1	(21)
		MCWB	66.1	65.6	64.1	60.7	55.5	51.5	50.5	54.1	57.5	59.7	62.9	64.9	(22)
	5%	DB	85.6	85.4	83.2	78.9	73.7	68.6	69.5	76.0	83.0	85.5	85.7	84.4	(23)
		MCWB	66.0	65.6	63.6	60.3	54.8	50.6	50.0	53.5	56.4	60.0	62.8	65.2	(24)
	10%	DB	82.8	82.7	80.5	75.9	71.2	66.1	66.4	73.0	80.0	82.1	82.6	82.1	(25)
		MCWB	65.9	65.7	63.9	60.0	54.1	50.0	49.1	52.4	55.5	59.6	62.4	65.3	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.0	70.9	70.2	66.6	60.9	56.5	56.1	60.2	62.9	66.4	68.9	70.3	(27)
		MCDB	82.1	81.3	78.9	77.3	71.0	69.1	68.0	75.2	78.9	79.5	81.9	81.5	(28)
	2%	WB	69.8	69.5	68.5	64.5	59.3	54.7	53.8	57.9	61.0	64.6	67.3	68.9	(29)
		MCDB	81.7	79.8	77.5	73.6	69.3	66.5	65.4	72.5	77.7	78.4	79.3	80.0	(30)
	5%	WB	68.7	68.4	67.3	63.0	57.9	53.3	52.1	55.9	59.4	63.4	66.1	67.7	(31)
		MCDB	79.8	78.3	76.4	72.5	68.1	64.2	64.8	70.1	75.8	77.3	77.6	78.7	(32)
	10%	WB	67.7	67.5	66.0	61.7	56.5	51.7	50.5	54.2	57.9	62.2	65.0	66.7	(33)
		MCDB	78.0	77.0	75.4	71.6	66.8	62.1	63.0	68.3	73.7	75.8	76.5	77.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.4	18.0	17.9	19.9	23.7	24.8	26.2	25.4	24.5	22.1	20.2	18.7	(35)
		MCDBR	22.9	22.3	22.7	24.5	27.1	28.6	30.3	29.2	28.2	26.7	24.7	22.3	(36)
	5% WB	MCWBR	7.9	7.2	7.6	8.9	10.7	12.4	13.4	11.7	10.3	8.9	7.9	7.9	(37)
		MCWBR	18.7	17.4	17.5	19.3	21.3	24.1	26.1	24.6	23.9	21.7	19.8	18.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.400	0.402	0.392	0.344	0.306	0.288	0.290	0.347	0.389	0.422	0.383	0.389	(40)
		taud	2.314	2.306	2.313	2.414	2.489	2.516	2.498	2.332	2.233	2.179	2.332	2.335	(41)
	Ebn at Noon		299	294	287	287	286	286	290	282	283	285	303	304	(42)
		Edh at Noon	44	43	41	34	29	26	28	36	44	49	43	43	(43)
(44) All-Sky Solar Radiation	RadAvg		2106	2001	1774	1519	1395	1266	1373	1616	1919	2063	2136	2148	(44)
	RadStd		214	179	164	135	70	64	34	73	88	136	196	126	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+1.19	+1.35	N/A	N/A	N/A	N/A	N/A	-326	+421	+157	

Nomenclature: See separate page